

DBMS – PRODUCT AND CATEGORY MANAGEMENT

BOOKSWAGON– E-Commerce Order Management Database System

1.DATABASE DESIGN:

1.1 CATEGORY TABLE :

Attribute	Description
Category_ID	Unique category ID
Category_Name	Category name

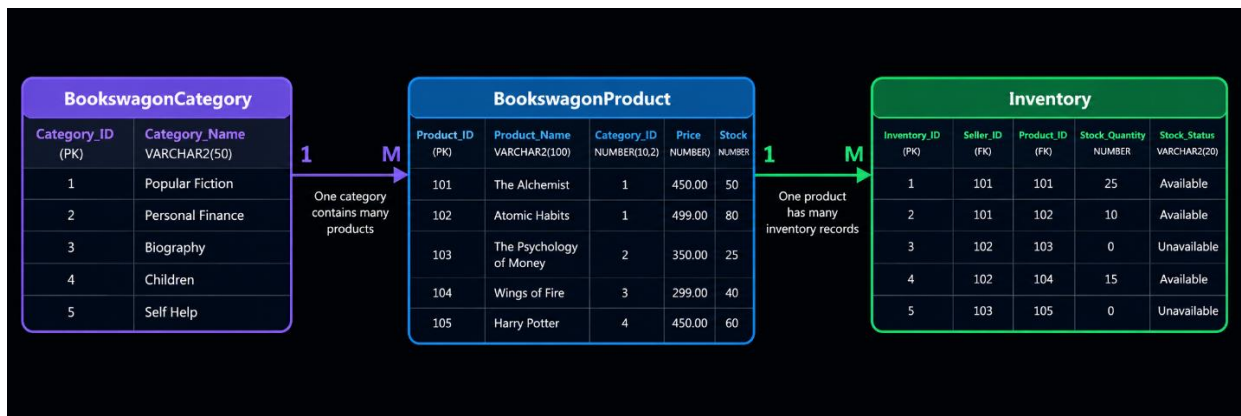
1.2 PRODUCT TABLE

Attribute	Description
Product_ID	Unique product ID
Product_Name	Product name
Price	Product price
Stock	Available stock
Category_ID	Category ID

2. PRIMARY AND FOREIGN KEY RELATIONSHIP

HuftCategory → HuftProduct

- BookswagonCategory.Category_ID → Primary Key
- BookswagonProduct.Product_ID → Primary Key
- BookswagonProduct.Category_ID → Foreign Key referencing BookswagonCategory.Category_ID
- Relationship: One-to-Many (1:M)



3.TABLE CREATION

3.1 CREATE CATEGORY TABLE

```
SQL> CREATE TABLE BookswagonCategory (  
2     Category_ID NUMBER PRIMARY KEY,  
3     Category_Name VARCHAR2(50)  
4 );
```

Table created.

3.2 CREATE PRODUCT TABLE

```
SQL> CREATE TABLE BookswagonProduct (  
2     Product_ID NUMBER PRIMARY KEY,  
3     Product_Name VARCHAR2(100),  
4     Category_ID NUMBER,  
5     Price NUMBER(10,2),  
6     Stock NUMBER,  
7     CONSTRAINT fk2_product_category  
8         FOREIGN KEY (Category_ID)  
9         REFERENCES BookswagonCategory(Category_ID)  
10 );
```

Table created.

4.DATA INSERTION

4.1 CATEGORY RECORDS

```
SQL> INSERT INTO BookswagonCategory VALUES (1, 'Fiction');
```

1 row created.

```
SQL> INSERT INTO BookswagonCategory VALUES (2, 'Finance');
```

1 row created.

```
SQL> INSERT INTO BookswagonCategory VALUES (3, 'Biography');
```

1 row created.

```
SQL> INSERT INTO BookswagonCategory VALUES (4, 'Children');
```

1 row created.

```
SQL> INSERT INTO BookswagonCategory VALUES (5, 'Self Help');
```

1 row created.

4.2 PRODUCT RECORDS

```
SQL> INSERT INTO BookswagonProduct VALUES (101, 'The Alchemist', 1, 399.00, 50);

1 row created.

SQL> INSERT INTO BookswagonProduct VALUES (102, 'Atomic Habits', 1, 499.00, 35);

1 row created.

SQL> INSERT INTO BookswagonProduct VALUES (103, 'The Psychology of Money', 2, 350.00, 25);

1 row created.

SQL> INSERT INTO BookswagonProduct VALUES (104, 'Wings of Fire', 3, 299.00, 40);

1 row created.

SQL> INSERT INTO BookswagonProduct VALUES (105, 'Harry Potter and the Philosopher Stone', 4, 450.00, 60);

1 row created.

SQL> INSERT INTO BookswagonProduct VALUES (106, 'Rich Dad Poor Dad', 2, 399.00, 30);

1 row created.
```

5.DATA RETRIVAL

5.1 DISPLAY CATEGORY RECORDS

```
SQL> SELECT * FROM BookswagonCategory;
```

CATEGORY_ID	CATEGORY_NAME
-------------	---------------

1	Fiction
2	Finance
3	Biography
4	Children
5	Self Help

5.2 DISPLAY PRODUCT RECORDS

```
SQL> SELECT*FROM Bookswagonproduct;
```

```
PRODUCT_ID
```

```
-----
```

```
PRODUCT_NAME
```

```
-----
```

```
CATEGORY_ID    PRICE    STOCK
```

```
-----
```

```
101
The Alchemist
1      399      50
```

```
102
Atomic Habits
1      499      35
```

```
PRODUCT_ID
```

```
-----
```

```
PRODUCT_NAME
```

```
-----
```

```
CATEGORY_ID    PRICE    STOCK
```

```
-----
```

```
103
The Psychology of Money
2      350      25
```

```
104
Wings of Fire
```

```
PRODUCT_ID
```

```
-----
```

```
PRODUCT_NAME
```

```
-----
```

```
CATEGORY_ID    PRICE    STOCK
```

```
-----
```

```
3      299      40
```

```
105
Harry Potter and the Philosopher Stone
4      450      60
```

```
106
```

```
PRODUCT_ID
```

```
-----
```

```
PRODUCT_NAME
```

```
-----
```

```
CATEGORY_ID    PRICE    STOCK
```

```
-----
```

```
Rich Dad Poor Dad
2      399      30
```

```
108
Think and Grow Rich
2      250      20
```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID    PRICE    STOCK
-----
      109
The Power of Your Subconscious Mind
      2      275      55

      110
Pride and Prejudice
      4      325      30

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID    PRICE    STOCK
-----

      111
The Hobbit
      4      499      70

10 rows selected.

```

6.PRODUCT UPDATE

```

SQL> UPDATE BookswagonProduct
  2  SET Price = 450
  3  WHERE Product_ID = 101;

1 row updated.

```

7.PRODUCT DELETION

```

SQL> DELETE FROM BookswagonProduct
  2  WHERE Category_ID = 5;

5 rows deleted.

```

8.CATEGORY UPDATE

```

SQL> UPDATE BookswagonCategory
  2  SET Category_Name = 'Popular Fiction'
  3  WHERE Category_ID = 1;

1 row updated.

SQL> UPDATE BookswagonCategory
  2  SET Category_Name = 'Personal Finance'
  3  WHERE Category_ID = 2;

1 row updated.

```

9. CATEGORY-WISE PRODUCT REPORT

```
SQL> SELECT
  2     c.Category_ID,
  3     c.Category_Name,
  4     p.Product_ID,
  5     p.Product_Name,
  6     p.Price,
  7     p.Stock
  8 FROM BookswagonCategory c
  9 JOIN BookswagonProduct p
 10 ON c.Category_ID = p.Category_ID
 11 ORDER BY c.Category_Name, p.Product_Name;
```

CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID

PRODUCT_NAME		

	PRICE	STOCK

	3 Biography	104
Wings of Fire	299	40

	4 Children	105
Harry Potter and the Philosopher Stone	450	60

CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID

PRODUCT_NAME		

	PRICE	STOCK

	4 Children	110
Pride and Prejudice	325	30

	4 Children	111
The Hobbit		

CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID

PRODUCT_NAME		

	PRICE	STOCK

	499	70
	2 Personal Finance	106
Rich Dad Poor Dad		
	399	30
	2 Personal Finance	109

CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID

PRODUCT_NAME		

	PRICE	STOCK

The Power of Your Subconscious Mind		
	275	55
	2 Personal Finance	103
The Psychology of Money		
	350	25

CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID

PRODUCT_NAME		

	PRICE	STOCK

	2 Personal Finance	108
Think and Grow Rich		
	250	20
	1 Popular Fiction	102
Atomic Habits		
	499	35

CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID

PRODUCT_NAME		

	PRICE	STOCK

	1 Popular Fiction	101
The Alchemist		
	399	50

10 rows selected.

10.CATEGORY-WISE TOTAL PRODUCTS REPORT

```
SQL> SELECT
  2     c.Category_ID,
  3     c.Category_Name,
  4     COUNT(p.Product_ID) AS Total_Products
  5 FROM BookswagonCategory c
  6 LEFT JOIN BookswagonProduct p
  7 ON c.Category_ID = p.Category_ID
  8 GROUP BY c.Category_ID, c.Category_Name
  9 ORDER BY c.Category_ID;
```

CATEGORY_ID	CATEGORY_NAME	TOTAL_PRODUCTS
1	Popular Fiction	2
2	Personal Finance	4
3	Biography	1
4	Children	3